Day 2 - Norms and Estimators

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From Exact to Approximate Fits

In the previous lecture, we saw that fitting a polynomial exactly through all data points leads to **overfitting**—the estimates become extremely sensitive to small changes in the data. The solution is to allow approximate fits, where our model doesn't pass through every point exactly.

Formally, instead of requiring $y_i = f(x_i)$, we introduce **residuals** $\epsilon_1, \epsilon_2, \dots, \epsilon_n$ and write

$$y_i = f(x_i) + \epsilon_i, \quad \text{for } i = 1, 2, \dots, n.$$

The residual ϵ_i measures how much the model prediction $f(x_i)$ differs from the observed value y_i. A good model should make the residuals "small"—but what exactly does "small" mean for a collection of n numbers?

This question leads us to a fundamental principle:

> **The Core Idea: Estimation = Minimization**
>
> Estimating a quantity from data is equivalent to minimizing the residuals with respect to a chosen notion of "size."
> Different notions of size lead to different estimators. This is the foundation of modern data science, machine learning, and AI. For example, training a neural network means choosing a loss function (a way to measure residual size) and minimizing it with respect to the network's parameters.

Today we'll explore this principle through the simplest possible estimation problem: estimating a constant from repeated measurements.

Estimating a Constant

Suppose you run an experiment to measure the gravitational acceleration g at Earth's surface. Due to measurement error, each trial gives a slightly different value. Here are 100 measurements (in m/s^2):

```
First 10 measurements: [ 9.986  9.85   9.908 10.034  9.997  9.712  9.905  9.795  9.8    9.85
Min: 9.555, Max: 10.037
```

What single value $\hat{g}$ best represents the "true" gravitational acceleration?

In estimation language, we want to find $\hat{g}$ such that

$$g_i = \hat{g} + \epsilon_i, \quad \text{for } i = 1, 2, \ldots, n, \tag{1}$$

where the residuals $\epsilon_i = g_i - \hat{g}$ are as "small" as possible.

> **Connection to Regression**
>
> This is actually a regression problem in disguise! We're fitting a "degree 0 polynomial" (a constant function) to the data. The techniques we develop here will generalize to fitting lines, parabolas, and more complex models.

Measuring Size: Norms

To make "small residuals" precise, we need to measure the size of the residual vector

$$\vec{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix} \in \mathbb{R}^n.$$

In mathematics, a function that measures vector size is called a **norm**.

Definition and Properties

A **norm** on $\mathbb{R}^n$ is a function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ satisfying:

1. **Positivity:** $\|\vec{x}\| \geq 0$ for all $\vec{x}$, with $\|\vec{x}\| = 0$ if and only if $\vec{x} = \vec{0}$.
2. **Scaling:** $\|\alpha\vec{x}\| = |\alpha| \cdot \|\vec{x}\|$ for all scalars α and vectors $\vec{x}$.
3. **Triangle inequality:** $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ for all $\vec{x}, \vec{y}$.

Why these axioms?

- **Positivity** is natural: size should be non-negative, and only the zero vector has zero size.
- **Scaling** says that stretching a vector by factor α stretches its size by $|\alpha|$.
- **Triangle inequality** is the key property that allows us to do analysis (calculus) with norms. It says the "direct path" is never longer than going through an intermediate point.

If we drop the requirement that $\|\vec{x}\| = 0 \Rightarrow \vec{x} = \vec{0}$, we get a **semi-norm**.

The L^p Family

The most important family of norms is the L^p **norms**, defined for $p \geq 1$:

$$\|\vec{x}\|_p = \left(|x_1|^p + |x_2|^p + \cdots + |x_n|^p\right)^{1/p}$$

Different values of p weight errors differently:

- **Small p:** treats all errors more equally
- **Large p:** increasingly dominated by the largest errors

Exercises: L^p Norms

Exercise 1: Prove that $\|\cdot\|_1$ satisfies all three norm axioms.
Exercise 2: Prove that $\|\cdot\|_2$ satisfies all three norm axioms.
Exercise 3: (Challenge) Prove that $\|\cdot\|_p$ is a norm for all $p \geq 1$. (Hint: Look up Minkowski's inequality.)

Three Estimators

We now apply three different L^p norms to our estimation problem and discover that each leads to a familiar statistic.

L^2 Norm: The Mean

The L^2 norm is the **Euclidean distance**:

$$\|\vec{\epsilon}\|_2 = \sqrt{\epsilon_1^2 + \epsilon_2^2 + \cdots + \epsilon_n^2}$$

Because squaring amplifies large values, the L^2 norm penalizes large errors heavily.

L^2 Estimation Problem

Find $\hat{g}$ that minimizes

$$\|\vec{\epsilon}\|_2 = \sqrt{(\hat{g} - g_1)^2 + (\hat{g} - g_2)^2 + \cdots + (\hat{g} - g_n)^2}$$

Exercises: L^2 Estimation

Exercise 4: Consider data $g_1 = 1, g_2 = 2, g_3 = 4, g_4 = 2$. Find the value of $\hat{g}$ that minimizes $\|\vec{\epsilon}\|_2$.
Exercise 5: For general data $g_1, g_2, \ldots, g_n$, find the minimizer $\hat{g}$. (Hint: It's easier to minimize $\|\vec{\epsilon}\|_2^2$ instead. Why does this give the same answer?)

L^1 Norm: The Median

The L^1 norm is the **Manhattan distance** (or taxicab norm):

$$\|\vec{\epsilon}\|_1 = |\epsilon_1| + |\epsilon_2| + \cdots + |\epsilon_n|$$

Unlike L^2, the L^1 norm treats all errors proportionally—a error twice as large contributes exactly twice as much to the total.

! L^1 Estimation Problem

Find $\hat{g}$ that minimizes

$$\|\vec{\epsilon}\|_1 = |\hat{g} - g_1| + |\hat{g} - g_2| + \cdots + |\hat{g} - g_n|$$

i Exercises: L^1 Estimation

Exercise 6: Consider data $g_1 = 1, g_2 = 2, g_3 = 4, g_4 = 2$. Find the value of $\hat{g}$ that minimizes $\|\vec{\epsilon}\|_1$. (Hint: Think geometrically—where should $\hat{g}$ be relative to the data points?)
Exercise 7: (Challenge) For general sorted data $g_1 \leq g_2 \leq \cdots \leq g_n$ with n odd, find the minimizer $\hat{g}$.
Exercise 8: (Challenge) For general sorted data $g_1 \leq g_2 \leq \cdots \leq g_n$ with n even, find the minimizer $\hat{g}$. (Careful: the answer may not be unique!)

L^∞ Norm: The Midrange

As $p \to \infty$, the L^p norm converges to the **maximum norm**:

$$\|\vec{\epsilon}\|_\infty = \max(|\epsilon_1|, |\epsilon_2|, \ldots, |\epsilon_n|)$$

This norm only cares about the single worst error—all other errors are irrelevant.

! L^∞ Estimation Problem

Find $\hat{g}$ that minimizes

$$\|\vec{\epsilon}\|_\infty = \max(|\hat{g} - g_1|, |\hat{g} - g_2|, \ldots, |\hat{g} - g_n|)$$

i Exercises: L^∞ Estimation

Exercise 9: (Challenge) Prove that $\lim_{p\to\infty} \|\vec{x}\|_p = \max(|x_1|, \ldots, |x_n|)$.
Exercise 10: Verify that $\|\cdot\|_\infty$ satisfies all three norm axioms.
Exercise 11: Consider data $g_1 = 1, g_2 = 2, g_3 = 4, g_4 = 2$. Find the value of $\hat{g}$ that minimizes $\|\vec{\epsilon}\|_\infty$.
Exercise 12: For general sorted data $g_1 \leq g_2 \leq \cdots \leq g_n$, find the minimizer $\hat{g}$.

Exercises: Geometry

Exercise 13: Plot the unit circles $\{\vec{x} \in \mathbb{R}^2 : \|\vec{x}\|_p = 1\}$ for $p = 1$, $p = 2$, and $p = \infty$. What shapes do you see?

- $p = 1$: $|x| + |y| = 1$
- $p = 2$: $x^2 + y^2 = 1$
- $p = \infty$: $\max(|x|, |y|) = 1$

Use Desmos to explore what happens as $p \to \infty$. What shape emerges? What happens for $p < 1$?

Summary: Three Norms, Three Statistics

We've discovered a beautiful unification: three familiar statistics arise from minimizing three different norms.

Main Result

Norm	Estimator $\hat{g}$	Common Name
L^2	$\frac{1}{n}\sum_{i=1}^{n} g_i$	**Mean**
L^1	middle value of sorted data	**Median**
L^∞	$\frac{\min(g_i)+\max(g_i)}{2}$	**Midrange**

Let's verify this with our gravitational acceleration data:

```
          Norm Estimate
     L² (mean)   9.8160
   L¹ (median)   9.8194
L∞ (midrange)   9.7958
    True value   9.8100
```

All three estimates are close to the true value, but they differ. Which should we prefer?

Each has different properties:

- **Mean (L^2):** Uses all data, but sensitive to outliers (one extreme value can pull it far off)
- **Median (L^1):** Robust to outliers (up to half the data can be corrupted!), but ignores magnitudes
- **Midrange (L^∞):** Only uses extreme values, extremely sensitive to outliers

The choice depends on what we know (or assume) about our measurement errors. This leads us to probability theory.

Looking Ahead

We've seen that different norms lead to different estimators. But we haven't answered the fundamental question: **which norm should we use?**

It turns out that the L^2 norm (and hence the mean) has a special status. In the next lecture, we'll introduce the **normal distribution** and see that:

1. If measurement errors follow a normal distribution, then the L^2 estimator is optimal in a precise sense (**maximum likelihood**).
2. This connection explains why the mean and the method of least squares are so ubiquitous in statistics.

We'll also see how this framework extends from estimating a single constant to fitting lines and more complex models—bringing us back to the regression problem from Day 1.